THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH2012 Fundamental Concepts of Mathematics
Solution to **Assignment 4**

1. (a) The relation R is an equivalence relation.
<u>R is reflexive:</u> For any $x \in \mathbb{Z}$, $2x + 2x = 4x \equiv 0 \pmod 4$, that is, xRx.
<u>R is symmetric:</u> For any $x, y \in \mathbb{Z}$, if xRy, that is, if $2x + 2y \equiv 0 \pmod 4$, then we have $2y + 2x \equiv 0 \pmod 4$ and hence yRx.
<u>R is transitive:</u> For any $x, y, z \in \mathbb{Z}$, if xRy and yRz, that is, if $2x+2y = 4k$ and $2y+2z = 4k'$ for some integers k, k', then we have $2x + 2z = 4(k + k' - y) \equiv 0 \pmod 4$ and hence xRz.
Therefore, R is an equivalence relation. To find all the distinct equivalence classes, let us begin with the equivalence class containing 0:

$$\begin{aligned}
[0] &= \{x \in \mathbb{Z} : xR0\} \\
&= \{x \in \mathbb{Z} : 2x \equiv 0 \pmod 4\} \\
&= \{x \in \mathbb{Z} : 2x = 4k \text{ for some integer } k\} \\
&= \{x \in \mathbb{Z} : x = 2k \text{ for some integer } k\} \\
&= \{x \in \mathbb{Z} : x \equiv 0 \pmod 2\}
\end{aligned}$$

Since the class $[0]$ does not consist of all integers, there should be at least one other equivalence class. Note that $1 \notin [0]$, we have $[1] \neq [0]$ and is disjoint from $[0]$. Then

$$\begin{aligned}
[1] &= \{x \in \mathbb{Z} : xR1\} \\
&= \{x \in \mathbb{Z} : 2x + 2 \equiv 0 \pmod 4\} \\
&= \{x \in \mathbb{Z} : 2x + 2 = 4k \text{ for some integer } k\} \\
&= \{x \in \mathbb{Z} : x = 2k - 1 \text{ for some integer } k\} \\
&= \{x \in \mathbb{Z} : x \equiv 1 \pmod 2\}.
\end{aligned}$$

Now since <u>every</u> integer belongs to <u>exactly one</u> of $[0]$ and $[1]$, so $[0]$ and $[1]$ are all the distinct equivalence classes.

(b) The relation R is an equivalence relation.
<u>R is reflexive</u>: For any $x \in \mathbb{Z}$, $4x + x = 5x \equiv 0 \pmod 5$, that is, xRx.
<u>R is symmetric</u>: For any $x, y \in \mathbb{Z}$, if xRy, that is, if $4x + y = 5k$ for some integer k, then we have $4y + x = 4(5k - 4x) + x = 5(4k - 3x) \equiv 0 \pmod 5$ and hence yRx.
<u>R is transitive</u>: For any $x, y, z \in \mathbb{Z}$, if xRy and yRz, that is, if $4x + y = 5k$ and $4y + z = 5k'$ for some integers k, k', then we have $4x + z = 5(k + k' - y) \equiv 0 \pmod 5$ and hence xRz.
Hence, R is an equivalence relation. Again, we begin with the equivalence class containing 0:

$$\begin{aligned}
[0] = \{x \in \mathbb{Z} : xR0\} &= \{x \in \mathbb{Z} : 0Rx\} \\
&= \{x \in \mathbb{Z} : x \equiv 0 \pmod 5\}
\end{aligned}$$

Since $1 \notin [0]$, we have $[1] \neq [0]$ and is disjoint from $[0]$. Then

$$\begin{aligned}
[1] \;=\; \{x \in \mathbb{Z} : xR1\} &= \{x \in \mathbb{Z} : 1Rx\} \\
&= \{x \in \mathbb{Z} : 4 + x \equiv 0 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv -4 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv 1 \pmod 5\}.
\end{aligned}$$

Since $2 \notin [0], [1]$, we have the class $[2]$ which is disjoint from $[0]$ and $[1]$. Then

$$\begin{aligned}
[2] \;=\; \{x \in \mathbb{Z} : xR2\} &= \{x \in \mathbb{Z} : 2Rx\} \\
&= \{x \in \mathbb{Z} : 8 + x \equiv 0 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv -8 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv 2 \pmod 5\}.
\end{aligned}$$

Since $3 \notin [0], [1], [2]$, we have the class $[3]$ which is disjoint from $[0], [1]$ and $[2]$. Then

$$\begin{aligned}
[3] \;=\; \{x \in \mathbb{Z} : xR3\} &= \{x \in \mathbb{Z} : 3Rx\} \\
&= \{x \in \mathbb{Z} : 12 + x \equiv 0 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv -12 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv 3 \pmod 5\}.
\end{aligned}$$

Since $4 \notin [0], [1], [2], [3]$, we have the class $[4]$ which is disjoint from $[0], [1], [2]$ and $[3]$. Then

$$\begin{aligned}
[4] \;=\; \{x \in \mathbb{Z} : xR4\} &= \{x \in \mathbb{Z} : 4Rx\} \\
&= \{x \in \mathbb{Z} : 16 + x \equiv 0 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv -16 \pmod 5\} \\
&= \{x \in \mathbb{Z} : x \equiv 4 \pmod 5\}.
\end{aligned}$$

Now since <u>every</u> integer belongs to <u>exactly one</u> of $[0], ..., [4]$, so $[0], ..., [4]$ are all the distinct equivalence classes.

(c) <u>R is reflexive</u>: For any $x \in \mathbb{R}$, $|x - x| = 0$, that is, xRx.
<u>R is symmetric</u>: For any $x, y \in \mathbb{R}$, suppose we have xRy, that is, $|x - y| \leq 1$. Then $|y - x| \leq 1$. Hence yRx.
<u>R is NOT transitive</u>: Take $x = 1$, $y = 2$ and $z = 3$. We have $1R2$ and $2R3$, but we do not have $1R3$ because $|1 - 3| > 1$.
Since this is not an equivalence relation, it fails to give a partition of $\mathbb{R}$.

(d) <u>R is NOT reflexive</u>: Take $x = 1$. Since $1 \cdot 1 = 1 \neq 0$, hence 1 is not related to 1 (any other nonzero real number x will also do the job).
<u>R is symmetric</u>: For any $x, y \in \mathbb{R}$, suppose we have xRy, that is, $xy = 0$. But $xy = yx$ for any $x, y \in \mathbb{R}$, so we have yRx.
<u>R is NOT transitive</u>: Take $x = 1$, $y = 0$ and $z = 2$. We have $1R0$ and $0R2$, but we do not have $1R2$ because $1 \cdot 2 \neq 0$
Since this is not an equivalence relation, it fails to give a partition of $\mathbb{R}$.

(e) <u>R is reflexive</u>: For any $x \in \mathbb{R}$, we can take $n = 1$ so that $x - 1 \cdot x = 0 \in \mathbb{Z}$. Thus we have xRx.
<u>R is NOT symmetric</u>: Take $x = 1 \in \mathbb{R}$ and $y = \frac{1}{2} \in \mathbb{R}$. Then we have xRy because we can choose $n = 2$ so that $1 - 2 \cdot \frac{1}{2} = 0 \in \mathbb{Z}$. However, for ANY $m \in \mathbb{Z}$, $\frac{1}{2} - m \cdot 1 = \frac{1}{2} - m$ is NOT an integer because if $\frac{1}{2} - m$ were an integer, then $\frac{1}{2}$ is also an integer as it is equal to the sum of two integers. This is a contradiction. It follows that $y = \frac{1}{2}$ cannot be related to $x = 1$.
<u>R is transitive</u>: For any $x, y, z \in \mathbb{R}$, if xRy and yRz, that is, if $x - n_1 y$ and $y - n_2 z$ are integers for some $n_1, n_2 \in \mathbb{Z}$, then

$$x - n_1 n_2 z = (x - n_1 y) + n_1(y - n_2 z) \quad \text{is also an integer.}$$

It follows that xRz because we are able to find an integer $n_1 n_2$ so that $x - n_1 n_2 z$ is an integer. Since this is not an equivalence relation, it fails to give a partition of $\mathbb{R}$.

2. (a) <u>R is NOT reflexive</u>: Take $S = \{1\}$ and $A = \emptyset$. Since $\emptyset \cup \emptyset = \emptyset \neq \{1\}$, so we do not have ARA.
   (b) <u>R is symmetric</u>: For any $A, B \in \mathcal{P}(S)$, suppose we have ARB, that is, $A \cup B = S$. But $A \cup B = B \cup A$ for any $A, B \in \mathcal{P}(S)$, so we have BRA.
   (c) <u>R is NOT transitive</u>: Take $S = \{1, 2\}$, $A = \{1\}$, $B = \{1, 2\}$ and $C = \emptyset$. We have ARB and BRC, but we do not have ARC.

3. <u>$R \cap S$ is reflexive</u>: We have xRx and xSx for all $x \in A$ since both R and S are reflexive. Now, for any $x \in A$,

$$(x, x) \in R \text{ and } (x, x) \in S \implies (x, x) \in R \cap S$$

Therefore, for all $x \in A$, we have $x(R \cap S)x$.

<u>$R \cap S$ is symmetric</u>: For any $x, y \in A$,

$$\begin{aligned}
x(R \cap S)y &\implies (x, y) \in R \cap S \\
&\implies (x, y) \in R \text{ and } (x, y) \in S \\
&\implies (y, x) \in R \text{ and } (y, x) \in S \quad \text{(since both } R \text{ and } S \text{ are symmetric)} \\
&\implies (y, x) \in R \cap S \\
&\implies y(R \cap S)x
\end{aligned}$$

Therefore, for any $x, y \in A$, we have $x(R \cap S)y \implies y(R \cap S)x$.

<u>$R \cap S$ is transitive</u>: For any $x, y, z \in A$, we have

$$((x, y) \in R \text{ and } (y, z) \in R) \implies (x, z) \in R$$

and

$$((x, y) \in S \text{ and } (y, z) \in S) \implies (x, z) \in S$$

since both R and S are transitive. Now, for any $x, y, z \in A$,

$$\begin{aligned}
x(R \cap S)y \text{ and } y(R \cap S)z &\implies (x, y) \in R \cap S \text{ and } (y, z) \in R \cap S \\
&\implies ((x, y) \in R \text{ and } (x, y) \in S) \text{ and } ((y, z) \in R \text{ and } (y, z) \in S) \\
&\implies ((x, y) \in R \text{ and } (y, z) \in R) \text{ and } ((x, y) \in S \text{ and } (y, z) \in S) \\
&\implies (x, z) \in R \text{ and } (x, z) \in S \quad \text{(by assumption)} \\
&\implies (x, z) \in R \cap S \\
&\implies x(R \cap S)z
\end{aligned}$$

Therefore, for any $x, y, z \in A$, we have $(x(R \cap S)y \text{ and } y(R \cap S)z) \implies x(R \cap S)z$.

4. $(\Longrightarrow)$:

For any $(x, y) \in R$, we have $(y, x) \in R$ since R is symmetric, which then implies $(x, y) \in R^{-1}$ by definition of the inverse relation. This proves $R \subseteq R^{-1}$.

On the other hand, for any $(x, y) \in R^{-1}$, we have $(y, x) \in R$ by definition of the inverse relation, which then implies $(x, y) \in R$ since R is symmetric. This proves $R^{-1} \subseteq R$.

$(\Longleftarrow)$:

For any $x, y \in A$, if $(x, y) \in R$, then $(x, y) \in R^{-1}$ since $R = R^{-1}$, which then implies $(y, x) \in R$ by definition of the inverse relation. Hence R is symmetric.

5. (a) $R_2 \circ R_1 \;=\; \{(1,9), (1,10), (2,8), (2,9), (2,10), (3,10), (4,8), (4,9), (4,10)\} \subseteq A \times C$.

   (b) First of all, both $R_3 \circ (R_2 \circ R_1)$ and $(R_3 \circ R_2) \circ R_1$ are subsets of $A \times D$. For any (a, d),

$$\begin{aligned}
(a,d) \in R_3 \circ (R_2 \circ R_1) \quad &\Longleftrightarrow \quad (a,c) \in (R_2 \circ R_1) \text{ and } (c,d) \in R_3 \text{ for some } c \in C \\
&\Longleftrightarrow \quad \big((a,b) \in R_1 \text{ and } (b,c) \in R_2\big) \text{ and } (c,d) \in R_3 \\
&\qquad\quad \text{for some } b \in B \text{ and } c \in C \\
&\Longleftrightarrow \quad (a,b) \in R_1 \text{ and } \big((b,c) \in R_2 \text{ and } (c,d) \in R_3\big) \\
&\qquad\quad \text{for some } b \in B \text{ and } c \in C \\
&\Longleftrightarrow \quad (a,b) \in R_1 \text{ and } (b,d) \in (R_3 \circ R_2) \text{ for some } b \in B \\
&\Longleftrightarrow \quad (a,d) \in (R_3 \circ R_2) \circ R_1.
\end{aligned}$$

   Hence we have $R_3 \circ (R_2 \circ R_1) \;=\; (R_3 \circ R_2) \circ R_1$.

6. <u>R is reflexive</u>: Since $a + b = b + a$, we have $(a, b)R(a, b)$ for any $(a, b) \in \mathbb{N} \times \mathbb{N}$.

   <u>R is symmetric</u>: For any $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$,

$$(a,b)R(c,d) \implies a + d = b + c \implies c + b = d + a \implies (c,d)R(a,b).$$

   <u>R is transitive</u>: For any $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$,

$$\begin{aligned}
(a,b)R(c,d) \text{ and } (c,d)R(e,f) \quad &\Longrightarrow \quad a + d = b + c \text{ and } c + f = d + e \\
&\Longrightarrow \quad a + d + c + f = b + c + d + e \\
&\Longrightarrow \quad a + f = b + e \\
&\Longrightarrow \quad (a,b)R(e,f).
\end{aligned}$$

7. (a) Since R is reflexive, for any $a \in S$, we have aRa, that is, $a - a = 0 \in S$.

   (b) For any $a \in \mathbb{Z}$,

$$a \in S \implies a - 0 \in S \implies aR0 \implies 0Ra \;\; (\text{as } R \text{ is symmetric}) \implies -a \in S.$$

   (c) For any $a, b \in \mathbb{Z}$,

$$\begin{aligned}
a, b \in S \implies a \in S \text{ and } -b \in S \;\; (\text{by (b)}) \quad &\Longrightarrow \quad aR0 \text{ and } (-b)R0 \\
&\Longrightarrow \quad aR0 \text{ and } 0R(-b) \;\; (\text{as } R \text{ is symmetric}) \\
&\Longrightarrow \quad aR(-b) \;\; (\text{as } R \text{ is transitive}) \\
&\Longrightarrow \quad a + b \in S.
\end{aligned}$$

8. (a) <u>R is reflexive</u>: For any $\ell \in S$, ℓ is parallel to itself. Hence, we have $\ell R \ell$.
<u>R is symmetric</u>: For any $\ell_1, \ell_2 \in S$, suppose we have $\ell_1 R \ell_2$, that is, ℓ_1 and ℓ_2 are parallel. Then we have $\ell_1 = \ell_2$ or they have no point in common. Either case implies that ℓ_2 and ℓ_1 are parallel and hence we get $\ell_2 R \ell_1$.
<u>R is transitive</u>: For any $\ell_1, \ell_2, \ell_3 \in S$, suppose we have $\ell_1 R \ell_2$ and $\ell_2 R \ell_3$. If $\ell_1 = \ell_3$, then we are done. Hence we may assume $\ell_1 \neq \ell_3$. We wish to show that ℓ_1 and ℓ_3 have no point in common. Assume to the contrary that P is a common point. Then ℓ_1 and ℓ_3 are both parallel to ℓ_2, and passing through P, which is impossible by the hint. We conclude that ℓ_1 and ℓ_3 have no point in common and hence we have $\ell_1 R \ell_3$.

(b) $[\ell] = \{\ell' \in S : \ell' R \ell\} = \{\ell' \in S : \ell' \text{ and } \ell \text{ are parallel}\} = \{\text{horizontal line } y = r : r \in \mathbb{R}\}$

(c) <u>Disproof</u>: Let ℓ_1 be the x-axis, ℓ_2 be the y-axis and ℓ_3 be the straight line $y = 1$. Then, we have $\ell_1 R' \ell_2$ and $\ell_2 R' \ell_3$, but we do not have $\ell_1 R' \ell_3$ since ℓ_1 and ℓ_3 do not intersect. So R' is not transitive and is therefore not an equivalence relation.

9. (a) <u>R is reflexive</u>: For any $(x, y) \in \mathbb{R}^2, y - x^2 = y - x^2$, hence $(x, y)R(x, y)$ so R is reflexive.
<u>R is symmetric</u>: For any $(x, y), (z, w) \in \mathbb{R}^2$, assume that $(x, y)R(z, w)$, that is, $y - x^2 = w - z^2$. Then $w - z^2 = y - x^2$ and hence $(z, w)R(x, y)$. So R is symmetric.
<u>R is transitive</u>: For any $(x, y), (z, w), (u, v) \in \mathbb{R}^2$, assume that $(x, y)R(z, w)$ and $(z, w)R(u, v)$, that is,

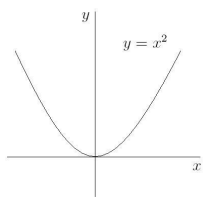
$$y - x^2 = w - z^2 \qquad \text{and} \qquad w - z^2 = v - u^2$$

Then $y - x^2 = w - z^2 = v - u^2$ and hence $(x, y)R(u, v)$. So R is transitive.

(b) The equivalence class $[(0, 0)]$ can be described as:

$$\left\{(x, y) \in \mathbb{R}^2 |\ y - x^2 = 0 - 0 = 0\right\} = \left\{(x, y) \in \mathbb{R}^2 |\ y = x^2\right\}$$

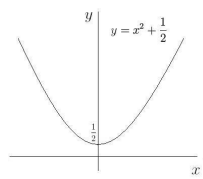
whose image is the parabola $y = x^2$:



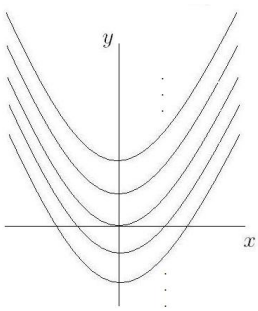
The equivalence class $[(1, 3/2)]$ can be described as:

$$\left\{(x, y) \in \mathbb{R}^2 |\ y - x^2 = \frac{3}{2} - 1^2 = \frac{1}{2}\right\} = \left\{(x, y) \in \mathbb{R}^2 |\ y = x^2 + \frac{1}{2}\right\}$$

whose image is the parabola $y = x^2 + 1/2$:



(c) After sketching a few equivalence classes, it should be clear that they partition $\mathbb{R}^2$ into a family of disjoint parabolae. Any given point $(x_0, y_0) \in \mathbb{R}^2$ lies in the class $[(0, y_0 - x_0^2)]$, and so any class is a translation of the parabola $y = x^2$ along the y-axis.



10. (a) <u>R is reflexive</u>: For any $(x, y) \in \mathbb{R}^2, x^2 + y^2 = x^2 + y^2$, hence $(x, y)R(x, y)$ so R is reflexive.
<u>R is symmetric</u>: For any $(x, y), (z, w) \in \mathbb{R}^2$, assume that $(x, y)R(z, w)$, that is, $x^2 + y^2 = z^2 + w^2$, then we also have $(z, w)R(x, y)$ and so R is symmetric.
<u>R is transitive</u>: For any $(x, y), (z, w), (u, v) \in \mathbb{R}^2$, assume that $(x, y)R(z, w)$ and $(z, w)R(u, v)$, that is,

$$x^2 + y^2 = z^2 + w^2 \qquad \text{and} \qquad z^2 + w^2 = u^2 + v^2$$

Then $x^2 + y^2 = u^2 + v^2$ and hence $(x, y)R(u, v)$. So R is transitive.

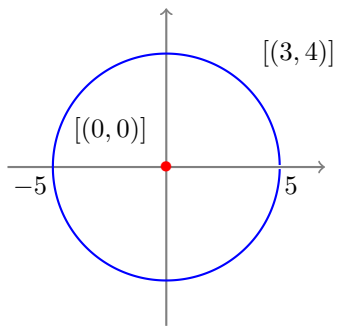
(b) The equivalence class $[(3, 4)]$ can be described as:

$$\left\{(x, y) \in \mathbb{R}^2 |\ (x, y)R(3, 4)\right\} = \left\{(x, y) \in \mathbb{R}^2 |\ x^2 + y^2 = 25\right\}$$

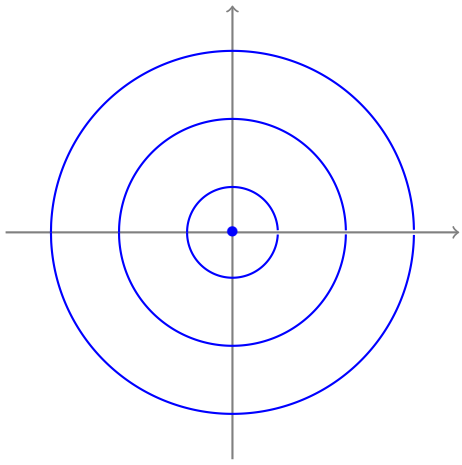
whose image is the circle centered at the origin with radius 5. The equivalence class $[(0, 0)]$ can be described as:

$$\left\{(x, y) \in \mathbb{R}^2 |\ (x, y)R(0, 0)\right\} = \left\{(x, y) \in \mathbb{R}^2 |\ x^2 + y^2 = 0\right\} = \{(0, 0)\},$$

which is just a point. See the following figure for reference:



(c) After sketching a few equivalence classes, it should be clear that these classes partition $\mathbb{R}^2$ into a family of disjoint circles and a single point:



11. (a) $(\Longrightarrow)$:

For any $n \in \mathbb{Z}$, suppose $n \not\equiv 1 \pmod 3$ and $n \not\equiv 2 \pmod 3$. This implies $n \equiv 0 \pmod 3$. Hence, we get $n^2 \equiv 0 \pmod 3$ from the hint, and so $n^2 \not\equiv 1 \pmod 3$.

$(\Longleftarrow)$:

For any $n \in \mathbb{Z}$, if $n \equiv 1 \pmod 3$ or $n \equiv 2 \pmod 3$, then $n = 3k+1$ for some $k \in \mathbb{Z}$ or $n = 3k'+2$ for some $k' \in \mathbb{Z}$. Thus we have $n^2 - 1 = (n-1)(n+1) = 3k(n+1)$ or $n^2 - 1 = (n-1)(n+1) = 3(n-1)(k'+1)$. In either case we conclude that $n^2 \equiv 1 \pmod 3$.

(b) <u>R is reflexive</u>: For any $x \in \mathbb{Z}$, $x^2 \equiv x^2 \pmod 3$, that is xRx.

<u>R is symmetric</u>: For any $x, y \in \mathbb{Z}$, if xRy, that is, $x^2 - y^2 = 3k$ for some integer k, then $y^2 - x^2 = -3k$, that is, $y^2 \equiv x^2 \pmod 3$. Hence we get yRx.

<u>R is transitive</u>: For any $x, y, z \in \mathbb{Z}$, if xRy and yRz, that is, $x^2 - y^2 = 3k$ for some integer k and $y^2 - z^2 = 3k'$ for some integer k', then $x^2 - z^2 = 3(k+k')$, that is, $x^2 \equiv z^2 \pmod 3$. Hence, we get xRz.

Therefore, R is an equivalence relation.

(c) We begin with the equivalence class (with respect to R) containing 0:

$$\begin{aligned} [0]_R = \{x \in \mathbb{Z} : xR0\} &= \left\{x \in \mathbb{Z} : x^2 \equiv 0 \pmod 3\right\} \\ &= \{x \in \mathbb{Z} : x \equiv 0 \pmod 3\} \quad \text{(by } \textit{Q.29(a) of Assignment 2}\text{)} \end{aligned}$$

Since $1 \notin [0]_R$, we know that the class $[1]_R$ must be disjoint from $[0]_R$. Then

$$\begin{aligned} [1]_R = \{x \in \mathbb{Z} : xR1\} &= \left\{x \in \mathbb{Z} : x^2 \equiv 1 \pmod 3\right\} \\ &= \{x \in \mathbb{Z} : x \equiv 1 \pmod 3 \text{ or } x \equiv 2 \pmod 3\}, \text{ by part (a).} \end{aligned}$$

Now note that $[0]_R = [0]_3$ and $[1]_R = [1]_3 \cup [2]_3$, where $[k]_3$ denotes the equivalence class of k with respect to the congruence of integers modulo 3, we conclude that <u>every</u> integer must belong to <u>exactly one</u> of $[0]_R$ and $[1]_R$. Hence $[0]_R$ and $[1]_R$ are all the distinct equivalence classes and so $\{[0]_R, [1]_R\}$ is a partition on $\mathbb{Z}$ with respect to the relation R.

12. We claim that $\{[0]_R, [1]_R, [2]_R\}$ gives a partition of $\mathbb{Z}$ with respect to R, where

$$\begin{aligned}
[0]_R &= \{x \in \mathbb{Z} : xR0\} = \{x \in \mathbb{Z} : x^2 \equiv 0^2 \pmod 5\} \\
[1]_R &= \{x \in \mathbb{Z} : xR1\} = \{x \in \mathbb{Z} : x^2 \equiv 1^2 \pmod 5\} \\
[2]_R &= \{x \in \mathbb{Z} : xR2\} = \{x \in \mathbb{Z} : x^2 \equiv 2^2 \pmod 5\}
\end{aligned}$$

From the hint, we have

$$\begin{aligned}
[0]_R &= \{x \in \mathbb{Z} : x \equiv 0 \pmod 5\} = [0]_5, \\
[1]_R &= \{x \in \mathbb{Z} : x \equiv 1 \pmod 5 \text{ or } x \equiv 4 \pmod 5\} = [1]_5 \cup [4]_5, \\
[2]_R &= \{x \in \mathbb{Z} : x \equiv 2 \pmod 5 \text{ or } x \equiv 3 \pmod 5\} = [2]_5 \cup [3]_5,
\end{aligned}$$

where $[k]_5$ denotes the equivalence class of k with respect to the congruence of integers modulo 5. We therefore conclude that <u>every</u> integer must belong to <u>exactly one</u> of $[0]_R$, $[1]_R$ and $[2]_R$, so $\{[0]_R, [1]_R, [2]_R\}$ gives a partition on $\mathbb{Z}$ with respect to the relation R.

13. (a) <u>R is reflexive</u>: For any $(x, y) \in \mathbb{Z} \times \mathbb{Z}$, $(x, y)R(x, y)$.
<u>R is symmetric</u>: For any $(x_1, y_1), (x_2, y_2) \in \mathbb{Z} \times \mathbb{Z}$, if $(x_1, y_1)R(x_2, y_2)$, that is, if $x_1 = x_2$, then we have $(x_2, y_2)R(x_1, y_1)$.
<u>R is transitive</u>: For any $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in \mathbb{Z} \times \mathbb{Z}$, if $(x_1, y_1)R(x_2, y_2)$ and $(x_2, y_2)R(x_3, y_3)$, that is, if $x_1 = x_2$ and $x_2 = x_3$, then $x_1 = x_3$, and hence we have $(x_1, y_1)R(x_3, y_3)$.

(b) The equivalence class $[(0, 0)]$ of $(0, 0)$ is the set $\{(0, y) : y \in \mathbb{Z}\}$. Since $1 \neq 0$, so $(0, 0)$ is not related to $(1, 0)$ and hence $[(0, 0)] \neq [(1, 0)]$.

(c) We begin with the equivalence class containing $(0, 0)$. By (b), $[(0, 0)] = \{(0, y) : y \in \mathbb{Z}\}$. Since $(1, 0) \notin [(0, 0)]$, we have the class $[(1, 0)] = \{(1, y) : y \in \mathbb{Z}\}$ which is disjoint from $[(0, 0)]$. Since $(2, 0) \notin [(0, 0)]$ and $[(1, 0)]$, we have the class $[(2, 0)] = \{(2, y) : y \in \mathbb{Z}\}$ which is disjoint from $[(0, 0)]$ and $[(1, 0)]$. Now we observe that there are infinitely many distinct equivalence classes with respect to this relation R, and we here list a few of them :

$$\cdots, [(-2, 0)], [(-1, 0)], [(0, 0)], [(1, 0)], [(2, 0)], \cdots$$

The equivalence classes can be parametrized by $[(x, 0)]$ for any $x \in \mathbb{Z}$.

14. (a) <u>R is reflexive</u>: For any $x \in \mathbb{R}$, we have $x - x = 0$ and $0 \in \mathbb{Z}$. Hence we have xRx.
<u>R is symmetric</u>: For any $x, y \in \mathbb{R}$, if xRy, that is, if $x - y = m$ for some $m \in \mathbb{Z}$, then we have $y - x = -m$ and $-m \in \mathbb{Z}$, and hence yRx.
<u>R is transitive</u>: For any $x, y, z \in \mathbb{R}$, if xRy and yRz, that is, if $x - y = m$ and $y - z = n$ for some $m, n \in \mathbb{Z}$, then we have $x - z = (x - y) + (y - z) = m + n$ and $m + n \in \mathbb{Z}$, and hence xRz.

Therefore, R is an equivalence relation.

(b) $[0] = \{x \in \mathbb{R} : xR0\} = \{x \in \mathbb{R} : x - 0 \in \mathbb{Z}\} = \mathbb{Z}$.

(c)

$$\begin{aligned}[\pi] \;=\; \{x \in \mathbb{R} : xR\pi\} \;=\; \{x \in \mathbb{R} : x - \pi \in \mathbb{Z}\} &= \{x \in \mathbb{R} : x - \pi = n \text{ for some } n \in \mathbb{Z}\} \\ &= \{x \in \mathbb{R} : x = \pi + n \text{ for some } n \in \mathbb{Z}\} \\ &= \{\pi + n : n \in \mathbb{Z}\}\end{aligned}$$

15. <u>Reflexive</u>: For any $x \in \mathbb{Z}$, we have $|x-2| = |x-2|$ and hence xRx is true.

    <u>Symmetric</u>: For any $x, y \in \mathbb{Z}$, if xRy, then $|x-2| = |y-2|$. But this also means yRx.

    <u>Transitive</u>: For any $x, y, z \in \mathbb{Z}$, if xRy and yRz, then $|x-2| = |y-2|$ and $|y-2| = |z-2|$. This implies $|x-2| = |z-2|$ and hence we have xRz.

    For any $n \in \mathbb{Z}$,

    $$[n] \;=\; \{x \in \mathbb{Z} : xRn\} \;=\; \{x \in \mathbb{Z} : |x-2| = |n-2|\}.$$

    Solving $|x-2| = |n-2|$ yields $x = n$ or $4 - n$. Hence the result.

    (*Remark*: Note that when $n = 2$, the class $[2] = \{2\}$ contains only one element.)

16. (a) <u>Reflexive</u>: For any $(x, y) \in \mathbb{R}^2$, we have $\max\{|x|, |y|\} = \max\{|x|, |y|\}$ and hence $(x, y)R(x, y)$ is true.

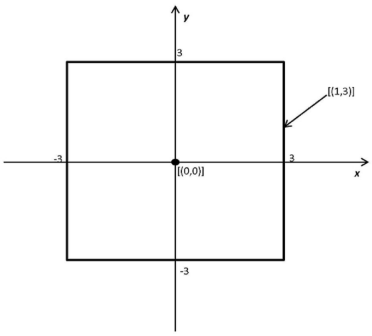
    <u>Symmetric</u>: For any $(x, y), (a, b) \in \mathbb{R}^2$, if $(x, y)R(a, b)$, then $\max\{|x|, |y|\} = \max\{|a|, |b|\}$. But this also means $(a, b)R(x, y)$.

    <u>Transitive</u>: For any $(x, y), (a, b), (c, d) \in \mathbb{R}^2$, if $(x, y)R(a, b)$ and $(a, b)R(c, d)$, then

    $$\max\{|x|, |y|\} \;=\; \max\{|a|, |b|\} \quad \text{and} \quad \max\{|a|, |b|\} \;=\; \max\{|c|, |d|\}.$$

    This implies $\max\{|x|, |y|\} = \max\{|c|, |d|\}$ and hence we have $(x, y)R(c, d)$.

    (b) See the following figure:



    We have $[(0, 0)] = \{(0, 0)\}$ and

    $$\begin{aligned}[(1, 3)] \;=\; &\{(x, 3) \in \mathbb{R}^2 : -3 \le x \le 3\} \cup \{(3, y) \in \mathbb{R}^2 : -3 \le y \le 3\} \\ &\cup \{(x, -3) \in \mathbb{R}^2 : -3 \le x \le 3\} \cup \{(-3, y) \in \mathbb{R}^2 : -3 \le y \le 3\}.\end{aligned}$$

    Put it in words, the class $[(0, 0)]$ is a set consisting only the origin, whereas the class $[(1, 3)]$ is the set of points on the <u>boundary</u> (not including the interior) of the square with length 6 shown in the above figure.

17. <u>Proof</u>:
For any $x, y \in A$, we know that either $[x] \cap [y] = \emptyset$ or $[x] \cap [y] \neq \emptyset$. If $[x] \cap [y] = \emptyset$, then we are done. If $[x] \cap [y] \neq \emptyset$, then there exists $u \in [x] \cap [y]$. This means $u \in [x]$ and $u \in [y]$. By definition, we have uRx and uRy. Since R is symmetric and transitive, we also have xRy. Now it follows from *Theorem 4.17* that $[x] = [y]$ and we are done.

18. (a) Here are all possible partitions of A:

$$P_1 = \{\{1\}, \{2\}, \{3\}\},\ P_2 = \{\{1, 2\}, \{3\}\},\ P_3 = \{\{1, 3\}, \{2\}\},\ P_4 = \{\{2, 3\}, \{1\}\},\ P_5 = \{\{1, 2, 3\}\}.$$

Here are the relations corresponding to the above partitions:

$$\begin{aligned}
R_1 &= \{(1,1), (2,2), (3,3)\} \\
R_2 &= \{(1,1), (2,2), (3,3), (1,2), (2,1)\} \\
R_3 &= \{(1,1), (2,2), (3,3), (1,3), (3,1)\} \\
R_4 &= \{(1,1), (2,2), (3,3), (2,3), (3,2)\} \\
R_5 &= \{(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}
\end{aligned}$$

(b) <u>R is reflexive</u>: For any $x \in A$, there exists a unique $S \in P$ such that $x \in S$ (by the definition of partition). Hence xRx and R is reflexive.
<u>R is symmetric</u>: For any $x, y \in A$, assume that xRy, that is, there exists $S \in P$ such that $x, y \in S$. Then we also have yRx and so R is symmetric.
<u>R is transitive</u>: For any $x, y, z \in A$, assume that xRy and yRz, that is, there exist $S, T \in P$ such that $x, y \in S$ and $y, z \in T$. According to the definition of partition P, y belongs to a unique member of P and hence we must have $S = T$. Therefore, $x, z \in S = T$, meaning that xRz and so R is transitive.